

Connection Between the Golden Ratio, Euler's Number, and π in Quantum Mechanics and Dynamical Systems

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1 Introduction

Three fundamental constants frequently appear in physics:

$$\text{Golden Ratio: } \phi = \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad (1)$$

$$\text{Euler's Number: } e \approx 2.718 \quad (2)$$

$$\text{Pi: } \pi \approx 3.14159 \quad (3)$$

These constants represent three fundamental physical processes:

- e controls exponential evolution
- π controls oscillations and rotations
- ϕ controls optimal stability and quasi-periodic structure

In quantum mechanics and dynamical systems, all three appear naturally in state evolution, interference, and stability.

2 Euler's Number and Quantum Time Evolution

The evolution of a quantum state is governed by:

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle \quad (4)$$

This equation uses Euler's number to describe continuous evolution.
Using Euler's formula:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (5)$$

we see that exponential evolution produces oscillations involving π .
Thus, quantum evolution directly connects e and π .

3 Role of π in Quantum Oscillations

Quantum systems behave like oscillators.

Energy eigenstates evolve as:

$$|\psi(t)\rangle = e^{-iEt/\hbar}|\psi(0)\rangle \quad (6)$$

If:

$$\omega = \frac{E}{\hbar} \quad (7)$$

then:

$$|\psi(t)\rangle = e^{-i\omega t}|\psi(0)\rangle \quad (8)$$

The oscillation period is:

$$T = \frac{2\pi}{\omega} \quad (9)$$

This shows π controls quantum oscillation cycles.

4 Golden Ratio and Quantum Stability

The Golden Ratio appears in quasi-periodic systems.

A quasi-periodic phase evolution can be written as:

$$\theta_n = 2\pi \frac{n}{\phi} \quad (10)$$

Because ϕ is irrational, this prevents resonance.

This improves stability in:

- Quantum coherence
- Quantum control sequences
- Dynamical stability

This is useful in quantum circuits.

5 Unified Expression Connecting e , π , and ϕ

Quantum evolution involves exponential rotation:

$$e^{i\omega t} \quad (11)$$

If we introduce Golden Ratio modulation:

$$\omega_n = \omega \phi^n \quad (12)$$

then evolution becomes:

$$|\psi(t)\rangle = e^{-i\phi^n \omega t} |\psi(0)\rangle \quad (13)$$

Since oscillations depend on π :

$$\omega = \frac{2\pi}{T} \quad (14)$$

we obtain:

$$|\psi(t)\rangle = e^{-i\phi^n (2\pi/T)t} |\psi(0)\rangle \quad (15)$$

This equation connects:

- Exponential evolution (e)
- Oscillation geometry (π)
- Stability scaling (ϕ)

6 Connection in Quantum Phase Space

Quantum states evolve in phase space:

$$\psi = e^{i\theta} \quad (16)$$

where phase rotation involves π .

If phase increments follow Golden Ratio spacing:

$$\theta_n = 2\pi \phi^{-n} \quad (17)$$

this produces optimal phase coverage.

This reduces interference instability.

7 Golden Ratio and Energy Level Structure

Some quantum systems have energy ratios approaching ϕ :

$$\frac{E_{n+1}}{E_n} \rightarrow \phi \quad (18)$$

This appears in:

- Quasi-crystals
- Fibonacci quantum chains

- Quantum chaos systems

Energy evolution still follows exponential form:

$$e^{-iEt/\hbar} \tag{19}$$

Thus linking all three constants.

8 Application in Quantum Circuits

Quantum gates use exponential rotations:

$$U = e^{-i\theta\sigma} \tag{20}$$

where:

$$\theta = \pi\phi^{-n} \tag{21}$$

This provides:

- Improved phase coverage
- Reduced resonance errors
- Better coherence stability

9 Dynamical Systems Interpretation

All three constants represent fundamental dynamical processes:

$$e \rightarrow \text{continuous evolution} \tag{22}$$

$$\pi \rightarrow \text{oscillatory geometry} \tag{23}$$

$$\phi \rightarrow \text{optimal stability} \tag{24}$$

Combined evolution:

$$\psi(t) = e^{-i(2\pi\phi^{-n})t} \tag{25}$$

This represents stable quantum oscillations.

10 Conclusion

The constants e , π , and ϕ form a natural mathematical framework for quantum mechanics and dynamical systems.

Their physical roles are:

- e : governs exponential quantum evolution
- π : governs oscillations and phase rotation
- ϕ : governs stability and optimal spacing

Together they describe stable, oscillatory quantum evolution and optimized quantum circuit behavior.

These constants provide a deep mathematical foundation for quantum dynamics, coherence, and stability.